FUNCTIONS & DEBUGGING

DAY 3

1. Simple Function – Sum

Problem:
Create a function sum(int a, int b) that returns sum of two numbers.

Input:
Two integers a b

Output:
Sum

---

2. Function with Return Type

Problem:
Create a function to return square of a number.

Input:
Integer n

Output:
Square of n

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3. Void Function

Problem:
Create a function that prints "Hello Java".

Input:
No input

Output:
Hello Java

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4. Maximum Using Function

Problem:
Write a function max(a, b) that returns the larger number.

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5. Even/Odd Function

Problem:
Write a function that takes a number and returns "Even" or "Odd".

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6. Factorial Using Function

Problem:
Create a function to compute factorial of a number.

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7. Prime Check Function

Problem:
Write a function isPrime(n) that returns true/false.

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8. Function Calling Function

Problem:
Create two functions:

- square(n)
- cube(n)
  Call both from main.

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9. Pass by Value Demo

Problem:
Write a function that tries to swap two numbers.
Print values before and after function call.

Goal:
Understand Java pass-by-value behavior

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10. Return Multiple Values

Problem:
Given a number, return:

- sum of digits
- product of digits

(Hint: use multiple function calls or custom logic)

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11. Function Overloading

Problem:
Create two functions:

- add(int a, int b)
- add(int a, int b, int c)

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12. Recursive Function

Problem:
Find factorial using recursion.

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13. Power Function

Problem:
Calculate a^b using recursion.

Debugging Based Questions

14. Find the Error

Problem:

```
int a = 10
System.out.println(a);
```

Task:
Identify and fix the error.

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15. Logical Error

Problem:

```
int a = 5, b = 10;
System.out.println(a - b);
```

Task:
Fix to print sum.

---

16. Runtime Error

Problem:

```
int a = 10, b = 0;
System.out.println(a / b);
```

Task:
Handle error safely.

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17. Wrong Output Bug

Problem:

```
int sum = 0;
for(int i = 1; i <= 5; i++);
    sum += i;
System.out.println(sum);
```

Task:
Fix logic error.

---

18. Function Bug

Problem:

```
int square(int n) {
    return n * n;
}
```

Called as:

```
System.out.println(square(5));
```

Task:
Fix missing parts.

Modular Coding Practice

19. Break into Functions

Problem:
Write program to:

- take input
- find sum of digits

- print result

Constraint:
Use 3 separate functions.

20. Reusable Function Design

Problem:
Create a function that:

- takes 2 numbers + operator
- returns result

(Reusable calculator function)

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21. Maximum of Two Numbers

Problem:
Create a method `max(int a, int b)` that returns the maximum of two numbers.

Input:
Two integers `a b`

Output:
Maximum value

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22. Number Type (Positive/Negative/Zero)

Problem:
Create a method `checkNumber(int n)` that returns:

- `"Positive"`
- `"Negative"`
- `"Zero"`

---

23. Even or Odd

Problem:
Create a method `isEven(int n)` that returns `true` if number is even, otherwise `false`.

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24. Leap Year

Problem:
Create a method `isLeapYear(int year)` that returns `true` if leap year, else `false`.

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25. Check Alphabet

Problem:
Create a method `isAlphabet(char c)` that returns `true` if character is alphabet.

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26. Vowel or Consonant

Problem:
Create a method `isVowel(char c)` that returns `true` if character is vowel.

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27. Uppercase or Lowercase

Problem:
Create a method `isUpperCase(char c)` that returns `true` if uppercase, else `false`.

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28. Even Numbers Array

Problem:
Create a method `getEvenNumbers()` that returns all even numbers from 1 to 100 as an array.

Output:
Array of even numbers

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29. Squares Array

Problem:
Create a method `getSquares()` that returns squares of numbers from 2 to 20.

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30. Cubes Array

Problem:
Create a method `getCubes()` that returns cubes of numbers from 2 to 20.
